

DAS732: Data Visualization / Term 1 (2026-27) / Programming Assignment 1 (A1)
International Institute of Information Technology Bangalore

Announcement Date: Aug 26, 2026 (Wed)

Last Date for Dataset Selection: Aug 30, 2026 (Sun)

Submission Deadline: 11:59 pm IST, Sep 13, 2026 (Sun)

Summary: Visual Exploration Using Visual Analytics Tools.

A1 tasks and requirements for a group assignment:

1. Read the selected dataset, process it, and explore the data using visualization and optionally other simple data analysis techniques, such as statistics.
2. Since it is visual exploration, multiple visualizations should collectively lead to knowledge discovery from the dataset.
 1. A data story must preferably evolve from multiple visualizations, instead of presenting several visualizations.
3. The assignment implementation must start with a question that can be answered by a list of visualization tasks implemented on the dataset. Visualization tasks include exploration and summary, along with descriptive verbs such as "overview", "trends", "search", etc. (If you are more interested in understanding tasks, please refer to Schulz et al. [1]). Your solution should solve n sets of tasks for an n-member team.
4. The visualizations may be generated on independent tools or an integrated tool, e.g., Tableau, Power BI, etc. We recommend the use of Tableau.
 1. You are advised not to split/fragment a task across different team members, as much as possible, and also ensure that all team members have tasks that will enable all in the team to make equal contributions.
 2. You may create tasks based on cohesive subsets of variables in the dataset.
5. The report shall contain a detailed description of the tasks you have tried and eventually implemented, who did what in the team, etc. Specifically, mention the contributions of each member.

A1 assessment:

Assessment is based on the report, images folder, and a video demo of the data stories.

1. The report should contain details on dataset description, tasks, visualizations, and inferences/conclusions. The report should have a section on Author Contributions where the contributions of the team members are explicitly mentioned.
2. The report should contain separate AI declaration forms for each team member.

3. Appropriate Python code or Tableau workbook shall be submitted. For the latter, appropriate access shall be managed for assessment.
4. The video demo should be crisp and at most 5 minutes long, where each member explains their task, visualization solutions, and inferences. The first minute can be used by the team leader or the data processing contributor to mention the preprocessing of the data.
5. The visualizations are expected to be simple plots that are understandable by laymen.
6. The folder of images must be organized properly, and all images in the folder must be featured in the report. The images in the folder may be named as Fig<number>.jpg/png/<image file extension>, where the number is the index of the image in the report. You may add additional images that are not contained in the report; in which case, include a readme file to indicate the significance of such images.

Notes:

1. If the submission is a Google Drive/OneDrive folder, the files shouldn't be edited after submission. If there is some editing done, then the date and time of submission will be recorded as the latest time stamp on any file or subfolder in the submitted folder.
2. It is the onus of the team members to ensure that the TAs and instructor have access to the submitted folder for grading until the end of the semester.

Rubrics for assessment:

The total points are out of 30, which will be scaled to 20 in the final grade.

- Choice of visualizations - 5 points
- Inferences - 5 points
- Video demo (including use of slides, clarity of presentation, Tableau demo) - 7.5 points
- Report (including good writing and presentation, the inclusion of images, captions, rationale on the choice of visualizations, inferences) - 7.5 points
- Completeness of submission (report, code, images, README) - 5 points

References:

[1] Schulz, H. J., Nocke, T., Heitzler, M., & Schumann, H. (2013). A design space of visualization tasks. *IEEE Transactions on Visualization and Computer Graphics*, 19(12), 2366-2375.